



A critical verification of beam and shell models of wind turbine blades

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ABSTRACT

Ever-increasing wind turbine size has challenged predictive capabilities on several fronts. To address part of the blade structural modeling uncertainty, a systematic model fidelity comparison study was conducted on commonly used finite elements. pyNuMAD was utilized to create beam, shell, and solid models of a 100 m long blade undergoing large static deflections. The solid model avoided the use of layered-solid elements by resolving core and facesheet layers. An unprecedented model with 73.7 million elements revealed insights that have never been possible from prior experimental and numerical studies. As compared to the solid element model, the tip deflection from the shell and beam model was found to be about 2% and 4.3% too low, respectively. The twist from the beam model was found to be about 5.6% too high, while the twist from shell model was 24% too low, though improvement was demonstrated with mesh refinement. The beam model adhesive stresses were more accurate than the shell model. Out-of-plane stresses were of great significance near geometric and material discontinuities, and neither the shell nor beam model captured these effects well. Failure predictions from beam, shell, or layered-solid models are unlikely to be reliable at trailing edges, adhesives, ply-drops, spar-cap boundaries.

1. Introduction

Electrical energy generation from renewable sources is crucial to reduce carbon emissions, and wind energy is a large source of existing and future capacity. Much of the success of wind energy over the years is partially due to steady increases in blade length which helps to drive energy costs down. Blade lengths of large utility-scale wind turbines has increased to point that they are the largest composite structures on Earth. With blade lengths approaching 120 m and beyond, design and analysis methods across the aero-structural-servo disciplines are reaching their limits [1]. Global responses such as dynamic instability calculations are increasingly less predictive. Local phenomenon such as laminate and adhesive fracture has recently led to millions of dollars in losses for both wind turbine manufacturers and owner/operators.

Beam models [2,3] for the blades are generally appropriate, since blades are long and slender. The position along the blade span is the only independent variable in the governing equations; thus, these are classified as 1D models. Not only are these widely used during preliminary design but also in numerous research activities. Wind blades are not only long and slender but also predominately thin-walled, so shell models [4,5] are also typically appropriate. While shell models are widely used, they are mostly restricted to detailed design considerations and some applications in fluid-structure-interaction research. With shell elements, the laminate thickness dimension is removed leaving

the two in-plane dimensions as independent variables, hence these are 2D models. They are better than beam models at accounting for geometric variations along the blade length but are poorer at capturing variations in a given cross section. The following paragraphs highlight the verification and validation deficiencies to both shell and beam models as well as how the present work fills many of the shortcomings.

Wind blade structural experiments provide many useful observations. Maximum sustained load, failure modes, deflections, and strains are all readily available to validate beam and shell models. The main limitation to existing experimental studies [6–8] is that only one longitudinal strain was considered and only at the outer surface along the spar caps. It is far less certain how well the remaining five strain components are predicted and how well the models predict strains though the laminate thicknesses, the adhesives and other parts of the blade. After all, more than one strain component is needed to predict failure. The deflection performance of beam and shell models are often undisputed. However, many questions remain on the torsional and edge-wise response. Both have become crucial as aeroelastic instabilities are a growing topic of concern [9].

Cauchy continuum models (referred to as “solid model” throughout) are the highest model fidelity possible but also the most difficult to solve. These are 3D models owing to the three spacial independent variables. Clearly, analytical solutions (eg [10]) to the governing equations of elasticity are not possible for wind blades due to enormous

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complexity of the micro- and mesoscale details as well as aerodynamic blade profiles. In the wind blade literature, computational and meshing effort is reduced by addressing the material heterogeneity through-the-thickness with so-called layered-solid elements. It is emphasized that these approximations are often accepted without question [11] and is the standard to which shell [6,8,12] and beam [13] models are held to. It is notable that Ref. [14] fully resolved the facesheet and core layers with solid elements, however, their geometry was not a wind blade but was a constant cross section abstraction with certain features of wind blades. Bertolini et al. [15] considered a blade-like tapered structure and highlighted some beam model stress deficiencies. In all of these cases, the geometry was oversimplified such that definite conclusions of model performance could not be carried to a full wind blade with confidence.

It is notable that the bulk of the advances to beam models comes from rotorcraft research [16–18]. They have been shown to work well for beams of constant cross section [19–23]. This is entirely typical of rotary wing aircraft, less so for wind blades. Beam models are currently unable to explicitly account for the span-wise geometric variations present in wind blades. The usual approach has been to mesh several cross sections along the blade span [24]. This approach has been shown to work well for blade deflections [25–27] but is far less certain how well these predict stresses. In other instances, the validity of beam models are established by making comparisons to shell models [25–28], despite their known issues. Furthermore, special provisions are made for beam models verification. Often the beams are isotropic [3] and sometimes anisotropic. They are mostly constant cross sections [3,13] with tip-loads.

This article presents a systematic comparison between beam (1D), shell (2D) and solid (3D) models of a full wind turbine blade under static loads. A unique aspect of this work is that the facesheet and core layers are fully resolved in the solid model. The resulting meshing difficulties are handled with pyNuMAD [29]. Solving the enormous solid model sizes were enabled by utilizing Sierra Solid Mechanics [30]. Most other model fidelity studies only consider global responses such as displacements and mode shapes. This work also compares the local stress responses at a variety of critical locations on the blade. The special provisions usually allowed for beam models is avoided to truly test its response.

2. Methods

In order to make objective comparisons between the different fidelities, it was important that the models be defined as geometrically and topologically equivalent as possible. Though some subtle geometric differences were inevitable, equivalence was maximized by generating all models using pyNuMAD [29], from the same IEA Wind Task 37 wind turbine ontology file [31].

Beam and shell models are routinely used to evaluate structural responses at large deflections of large blades. To fully test these models, a large blade capable of withstanding large deflections without buckling was desired. The BAR URC (Big Adaptive Rotor Upwind–Rail transportable–Carbon) [32] blade was selected since it is an open-source, 100 m long blade that had buckling considered during its design. All models were linear-elastic and geometrically nonlinear.

Cross sectional resultants from aero-servo-elastic simulations using beam models are typically transferred to shell or solid models [33,34]. This is because it is usually infeasible to perform those computations in the time-domain with shell or solid blade models. Since the current objective is to evaluate the accuracy of beam and shell models against a solid model, loads data will be transferred from the solid model to the beam model. Given the objective of comparing structural models of different fidelity, it was sufficient to choose a convenient loading scheme; so long as it is sufficiently realistic and static equivalence is achieved among the considered models. Choosing concentrated forces/moments

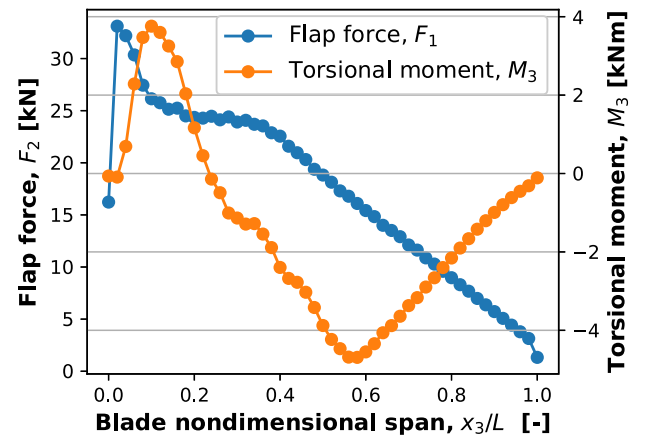


Fig. 1. Resulting nonzero point forces and moments as applied to beam model.

at various cross sections was avoided since different codes make different assumptions about how the forces are distributed to each of the nodes. The selected load type was a uniform traction distributed on the entire outer surface of the blade. Care was taken to ensure that each model used non-follower forces. The use of follower forces would cause the applied loading to be deflection dependant and would needlessly complicate model comparisons.

2.1. Beam model

The beam model was a geometrically exact Timoshenko model as implemented in BeamDyn [35]. Legendre spectral finite elements with Gaussian quadrature were used. According the Chen et al. [23], not all tools that compute beam properties are reliable. To provide the best possible result from the beam model fidelity, VABS [36] 4.0 was utilized to obtain the stiffness and mass matrices. Thirty cross-sections were meshed with pyNuMAD as described in Ref. [37] at 30 positions along the blade length.

VABS was also utilized to recover the point-wise stresses, strains, and displacements with its recovery capability. Stress resultants (three forces and three moments) in the deformed coordinates were requested in BeamDyn at every node location. These resultants were linearly interpolated to the corresponding locations of the cross sections used by VABS.

Uniform traction loading was directly achievable by the solid and shell models. However, the reduced order nature of beam models as currently formulated does not allow this loading. Instead, nodal force data was extracted from the solid model and applied to the beam model as numerous concentrated forces and moments along the reference line. First, a uniform traction that was small enough to produce little to no displacement was solved (1 Pa was convenient) with the solid model. Then, the reaction forces were subtracted from the external force data. The nodal forces were binned with a predetermined bin width, b . The remaining nodal forces in each bin were summed in all three global coordinate directions and applied to the beam model at the middle of the bin. The resulting torsional moment about the reference axis was also computed at each bin. The bins at the root and tip were $b/2$ wide. The resultants from these nodal forces were placed at $x_3 = 0$ and $x_3 = L$ in the Beam Model, where L is the blade length and was 100 m. A convergence analysis showed that $b = 2$ m (51 resultant locations) produced tip deflections within 0.1% of the case with $b = 0.5$ m (201 resultant locations). Thus $b = 2$ m was selected. The resulting force and moment distributions are shown in Fig. 1.

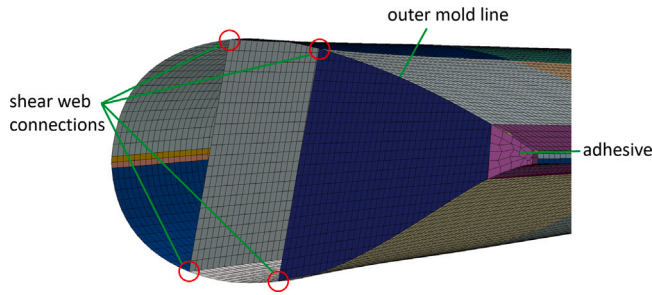


Fig. 2. Shell model cutaway illustrating its cross sectional definition.

2.2. Shell model

The shell model was constructed using the built-in mesh/model generation tools within pyNuMAD. The model data containing airfoils defining the outer mold line of the blade, the span-wise distribution of chord, sweep and twist and the material layup information was read from a file following the IEA Wind ontology [31]. The outer surface of the blade was defined by creating a series of smooth 3D splines connecting corresponding points between span-wise cross-sections. The splines were overlaid with an array of curved quadrilateral shell patches, each representing a certain section at a certain station of the blade. By default, each patch was meshed with a semi-structured grid of predominantly 4-node shell elements, with occasional 3-node elements to preserve approximately uniform element size. In close examination of torsional response as it relates to mesh refinement, the settings were modified to preserve a certain number of elements around the circumference of the blade all along its length. The shell section offsets were defined so that the reference surface formed by the elements represents outer mold line of the blade.

Shear webs were defined by connecting the suction side of the blade reference surface to the pressure side with a series of curved quadrilateral patches anchored on either side of the spar caps and running down the length of the blade. Trailing-edge adhesive was modeled by filling the space between the trailing-edge reinforcement on the pressure and suction side with solid elements, 8-node dominated with occasional 6-node triangular prism elements. The adhesive was defined as a separate part and attached to the outer surface shell structure with tie constraints. Fig. 2 illustrates the blade cross sectional structure as defined by the shell model.

Each section of the shell blade model was assigned material layup properties based on the specified in the original ontology file. The whole finite element model was written to file and imported into Abaqus for analysis. For the present study, a constant surface traction was applied to the blade's outer surface manually in the Abaqus user interface, and a geometrically nonlinear analysis was performed to determine the response. Abaqus S4 and S3 elements were utilized for all but the adhesive. The adhesive was comprised of C3D8 and C3D6 elements.

2.3. Solid model

Cubit was also utilized to create the solid model (continuum-solid elements). The identical cross sections that were created for the Beam Model were connected with span-wise splines. This time, the aerodynamic twist was left intact. A significant challenge was handling the transition between two main types of trailing-edge adhesive geometries. Near the root of most blades, the TE angles are large. This often necessitates the blade halves to be joined by adhesive that is in contact with the laminate thicknesses as shown Fig. 3(a)–(b). This is what is referred to here as Round TE adhesive. Moving towards the tip results in smaller TE angles as well as closer proximity of the

blade halves. This facilitates adhering the blade halves, not by laminate thickness, but with a lap-joint as illustrated in Fig. 3(f). This adhesive is referred to here as Flatback adhesive. Fig. 3(c)–(e) illustrate typical transition TEs used here. Fig. 3(c) illustrates the last TE adhesive that can be modeled as Round since the angle is getting to be smaller than a set threshold (50°). The next several TEs look like Fig. 3(d). The connection geometry is still considered as Round but the adhesive material properties are replaced with Glass Triax since the TE angle is too large for Flatback adhesive and the distance between the two blade halves is too large for Round adhesive. Once the blade halves are within 2.5% of the target adhesive thickness and the angle is less than another threshold (10°), both Round and Flatback adhesive geometries are included in the same cross section as shown in Fig. 3(e). All the Round TE adhesive geometries from this point back to the root (whether it has adhesive or Glass Triax properties) are connected to form volumes. The Flatback adhesive from this point on towards the blade tip are also connected. Once all volumes were created, they were meshed with 8-node hexahedral elements that had a Mean Quadrature formulation and a Midpoint strain incrimination. See Ref. [30] for further element details.

2.4. Notable geometric distinctions

While the greatest possible equivalence between the model geometries of the three fidelities was pursued, some subtle differences are worth noting. Many of these differences are inherent to each model fidelity. It is precisely these differences that we seek to quantify as a result of this study.

The geometry of the shell model differed from that of the solid model in three main ways. First, span-wise and chord-wise layer thickness transitions occurred as discontinuous steps. Individual sections of the shell model were treated as having constant thickness at any given spanwise location. The solid model addressed span-wise and chord-wise thickness transitions with continuity. Second, the shear webs in the shell model attached directly the shell reference surface which is the outer-mold-line of the blade. In the solid (and beam) model, the shear web is attached on inner surfaces of the blade. Thus for the shell model, the web layers partially overlap with the spar cap and panels layers, and there is no physical interaction at the inner surface of the blade. Furthermore, the shell model in pyNuMAD omits accounting for any web adhesives; the solid (and beam) model connects the web with adhesive at the inner surface. Third, like the shear webs, the trailing-edge adhesive in the shell model filled the total space between the outer surface on the suction and pressure side, effectively overlapping the reinforcement layers on the interior. The solid model accounted for the thickness of the reinforcement as shown in Fig. 3, filling the remaining space with adhesive.

As previously described, the beam model used identical cross-section definitions as the solid model. Thus, chord-wise thickness transitions and all adhesives are identical at a given cross section. However, the beam model formulation prohibits explicitly capturing span-wise geometric transitions. Span-wise geometric variation is handled indirectly in BeamDyn as it interpolates between defined cross sectional stiffness (and mass) properties.

2.5. Data extraction

The dimensional reductions of the beam and shell models created unique challenges in how stresses and displacements were extracted. On one hand, it was necessary to ensure that results being compared between the different models were taken from the same locations on the blade. But the points for which results are both readily accessible and accurate are different for each model. It was therefore necessary to implement an interpolation strategy to obtain the necessary data from each model.

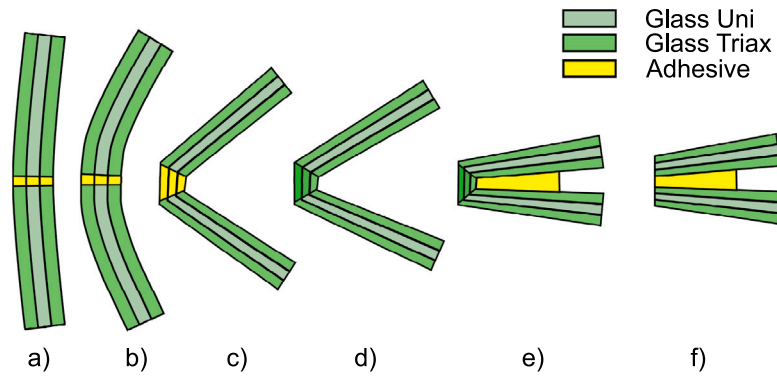


Fig. 3. Evolution of trailing-edge details for beam and solid models.

First a number of paths were defined where stresses and displacements would be queried. A “path” refers a sequence of coordinate points contained in the structure of the wind blade. All selected path points coincided with an integration point within one of the spanwise cross sections of the beam model. Some paths ran along the blade span (multiple beam cross sections) others progress through laminate thicknesses.

The use of Sierra Solid Mechanics [30] allowed for convenient interpolation of the stress and displacement data onto any coordinate points contained in the meshed domain. As such, element shape functions were used in the interpolation of the stress and displacement results onto the coordinate points for the solid model. Since the stress data was located on the integration points, it first had to be extrapolated to the nodes. This extrapolation was restricted to each material so that, at material interfaces, the adjacent material would not influence the results.

For the shell model, displacement results were reported at the element nodes, while stresses, strains, section strains and curvatures were reported at the element centroids and exported to file. These results were interpolated to the desired path points through general 3D linear interpolation from the nearest four points. For through-thickness paths, stresses and strains were obtained through the assumptions of composite laminate theory, using the reference surface strains and curvatures and the normal projection of each point from the reference surface since these points were not available directly from the results database.

For the beam model, integration point data was used directly for the beam model stress results. The displacements were obtained with a general 3D linear interpolation from the nearest four cross sectional nodes.

3. Results

Before completing the analysis on the full wind turbine blade, various preliminary studies were conducted in order to gain confidence in the methodology. These are collected in Section 3.1. Finally, model fidelity study results for the full wind turbine blade are presented in 3.2.

3.1. Preliminary studies

3.1.1. Isotropic and prismatic tube: D-tube

A simple geometry was sought that would test and refine our methodology. The decided problem was a constant cross section, 100 m long cantilevered tube. The cross section was semicircular with a web. This formed the shape of the letter “D” and, thus, this analysis case is referred to as “D-Tube”. The outer diameter was 4.5 m and each wall thickness was 80 mm. The material was all steel. The load for the shell and solid model was a uniform traction of 63 kPa on the semicircular portion of the cross section.

Fig. 4 shows the main deflection component as well as the predominant normal stress along a span-wise path on the compressing side of the tube. The Figure shows that satisfactory agreement was achieved among the models. It is notable that only the solid model shows a stress concentration near the clamped end.

3.1.2. Convergence studies

In order to ensure a fair comparison among each model, various convergence studies were required. The mesh convergence study on the solid model was performed with the first 5 stations, resulting in a blade root model that was 13.8 m long. A uniform flap-wise traction of 150 kPa was applied to the outer-mold-line. The convergence study was conducted by varying the number elements through the facesheet thicknesses. The through-thickness element edge lengths were then computed by dividing the layer thickness by the specified number through-thickness elements. The in-plane element size was kept to approximately 2.5 times the thickness dimension. This was well within generally accepted element aspect ratios. Note that the core layers are generally much larger than the facesheets but they had similar through-thickness element edge lengths. This resulted in finer mesh resolution in the core than in the facesheets. Also note that the adhesive layers had an equivalent number of elements through the thicknesses as the facesheets. Table 1 shows the mesh naming convention as well as the total number elements in each case.

Figs. 5 to 7 show the convergence behavior along selected paths for the displacements and stresses, respectively. Qualitative inspection of these plots shows that Coarse mesh is not converged. Given that the Fine mesh is over 3 times the size of the Medium mesh and with little change between the two, the Medium mesh is acceptable. Thus, the Medium mesh parameters (two elements through the facesheet thicknesses, aspect ratios of nominally 2.5) are used in creating the mesh of the full blade and thus the solid model results below.

Once the solid element mesh resolution was selected from the partial blade, the nonlinear solver convergence criteria for the full blade was studied. The full blade mesh at the selected resolution resulted 73,793,063 elements and 80,172,217 nodes. Results from the three relative residuals, r , that were studied are shown in Table 2. Successively smaller target relative residuals caused solver convergence difficulties. Thus the total time domain was divided into an increasing number of time steps as indicated in Table 2. Solid model results in the remaining sections are from the $r = 2 \times 10^{-3}$ relative residual as it was considered to be fully converged. It is remarkable that the coarsest case of $r = 2 \times 10^{-2}$ actually took longer to solve than the intermediate case of 2×10^{-3} . Clearly, $r = 2 \times 10^{-3}$ is optimal in terms of accuracy and speed. Note that the mesh sizing study on the partial blade was conducted at an even finer relative residual of $r = 2 \times 10^{-5}$.

The beam model formulation transforms a 3D problem to 2D cross sectional analyses and a 1D analysis of the reference line. This dimensional reduction necessitates the studying the convergence of (1)

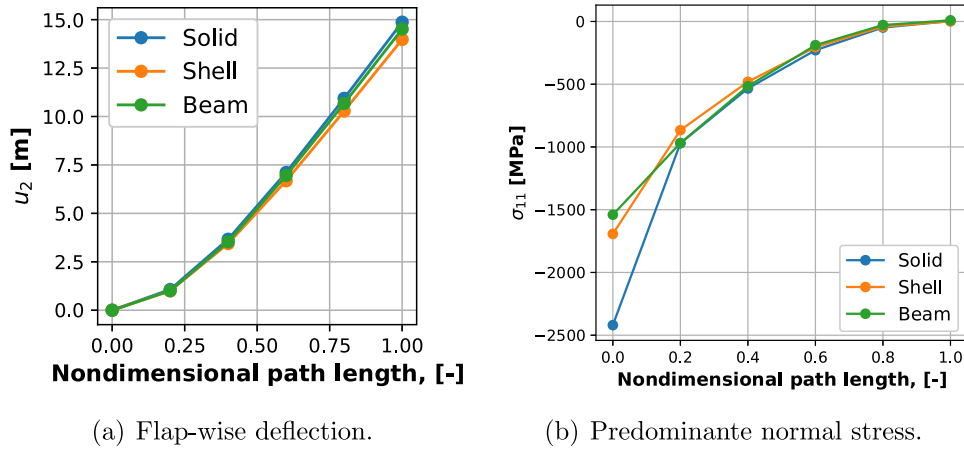


Fig. 4. Main deflection and stress from a preliminary analysis of a simple tube.

Table 1
Mesh sizing for solid model convergence study.

Mesh	Number of facesheet thickness elements	Total number of elements
Coarse	1	1,405,288
Medium	2	3,569,218
Fine	3	11,640,776

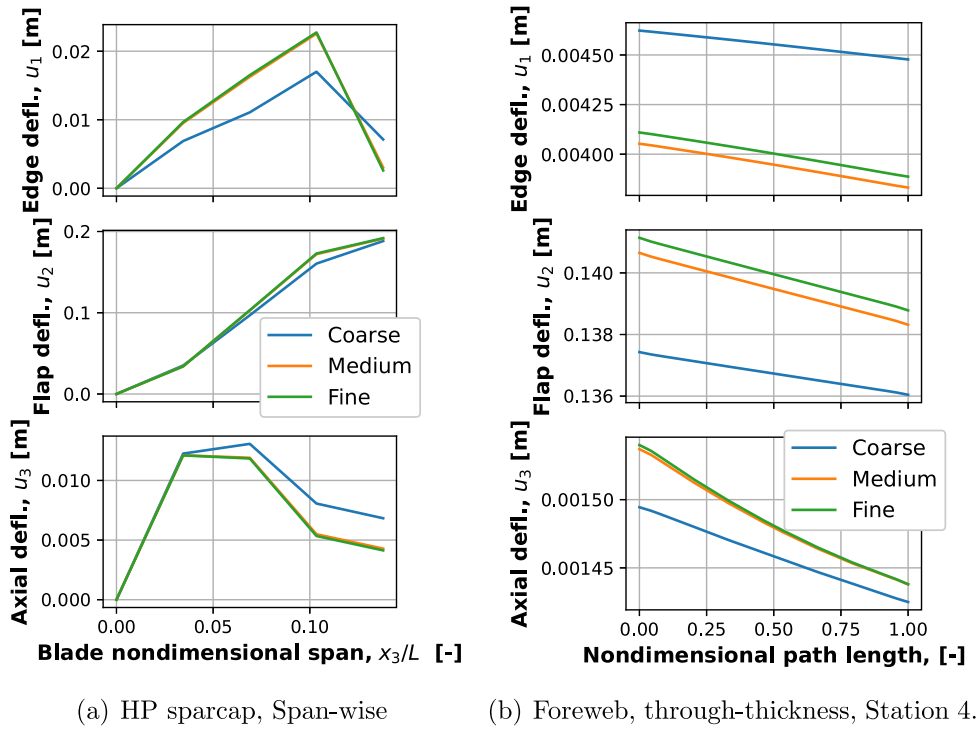


Fig. 5. Solid model convergence study – displacements.

Table 2
Relative residual study for solid model nonlinear solver on 2000 CPUs.

Relative residual, r	Number of time steps	Normalized tip defl.	Normalized Spar stress	Solve time [days]
2×10^{-2}	200	0.78624724	0.80097749	2.1
2×10^{-3}	500	0.99997631	0.99999125	1.4
2×10^{-4}	5000	1.0	1.0	4.7

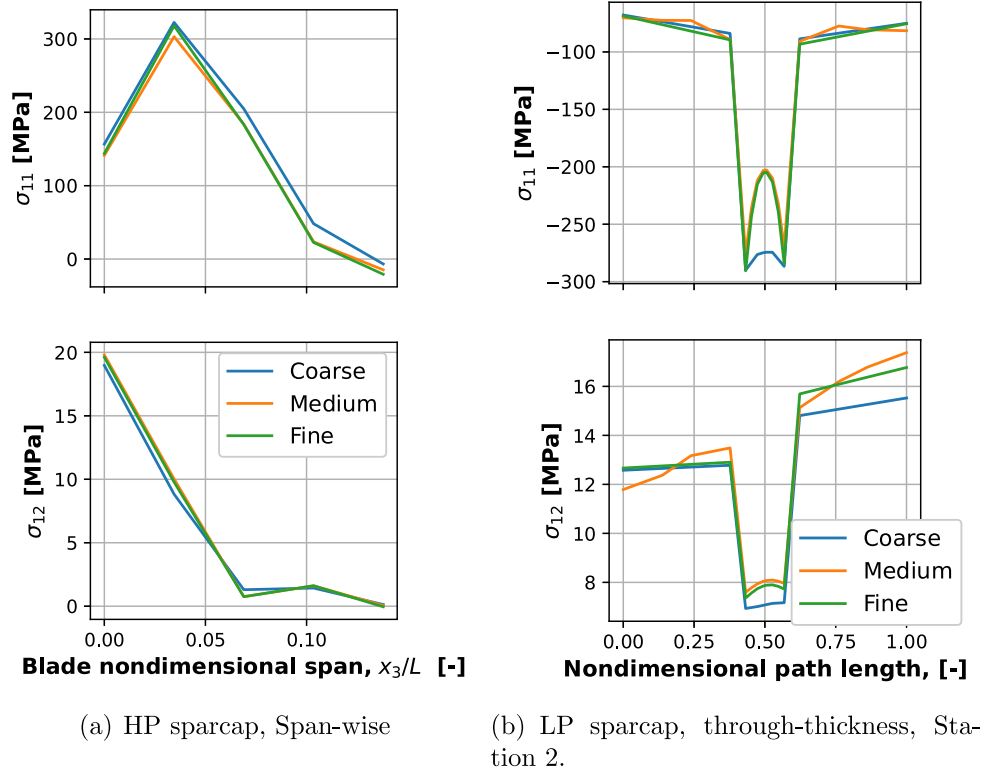


Fig. 6. Solid model convergence study – sparcap stresses. σ_{11} – Predominate normal stress. σ_{12} – in-plane shear stress.

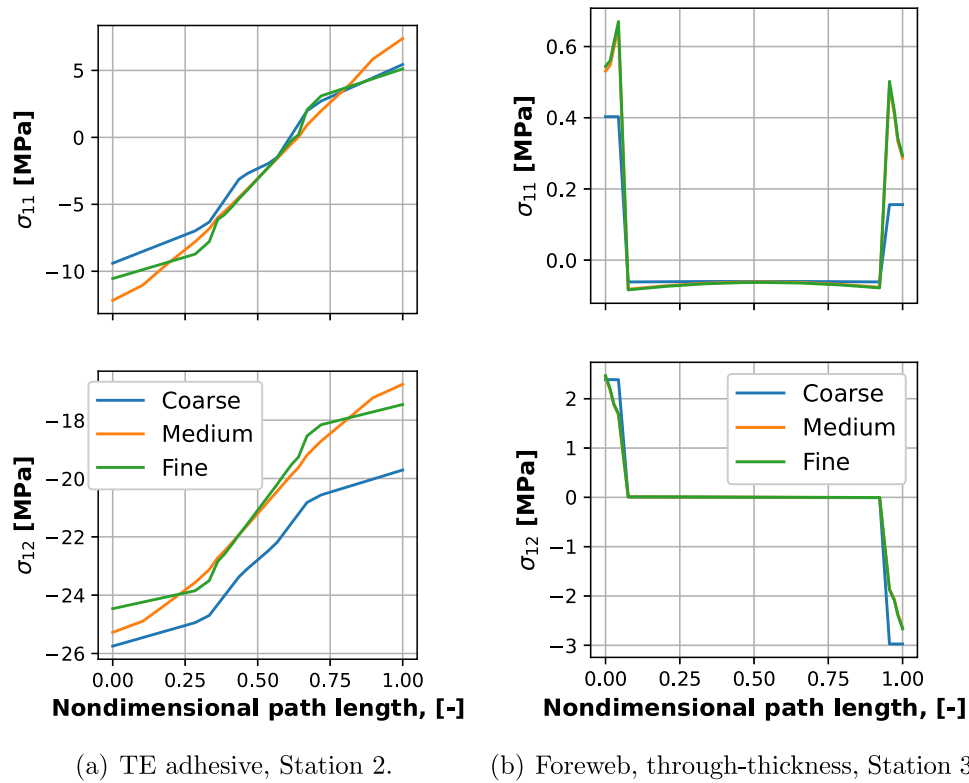


Fig. 7. Solid model convergence study – adhesive and web stresses. σ_{11} – Predominate normal stress. σ_{12} – in-plane shear stress.

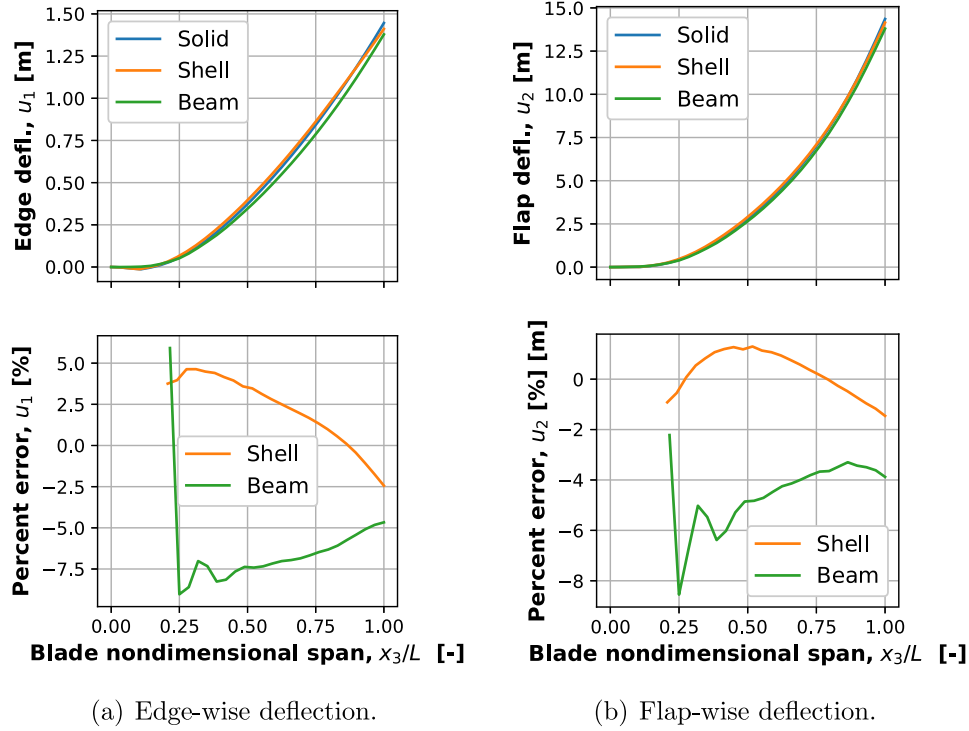


Fig. 8. Flap- and edge-wise deflections of the full wind turbine blade.

the cross sectional meshes, (2) BeamDyn mesh and (3) the approximation of the uniform traction as a series of concentrated forces and moments. For brevity, the discovered convergence characteristics are summarized. The cross sectional mesh resolution was studied on the D-Tube (Section 3.1.1). The meshes were created with the same codes as the wind blade. As such, three main layers were made where the thicknesses were to resemble that of sandwich construction of wind blades. Each layer was made of steel. Convergence was achieved rapidly with as little as one element in the facesheet thicknesses and a nominal aspect ratio of 5. This was expected since VABS is a semi-analytical method which converges faster than typical finite element methods. The selected mesh parameters for the beam model of the blade were two elements through the facesheets (and adhesives) and an aspect ratio of 2.5. These correspond with what was chosen for the solid model. Since BeamDyn uses high order elements, the order of the element interpolation function was studied. A 10th order element was found to be sufficient. Finally, the convergence study on the loads was described in Section 2.1

3.2. Complete wind blade results

3.2.1. Deflections

The deflection of a span-wise path in the high pressure spar cap is presented in Fig. 8. The edge-wise deflection is shown in Fig. 8(a) while the flap-wise deflection is shown in Fig. 8(b). Note, too, that the percent error relative to the solid model is also plotted. Division by very small numbers near the root caused plotting difficulties so the percent error data was truncated until the 7th station. In general, the beam and shell models both underpredict the edge-wise and flap-wise tip deflections; the beam model more so than the shell model. Edge-wise tip deflection errors are -2.4% for the shell model and -4.7% for the beam model. Flap-wise tip deflection errors are -1.5% for the shell model -3.9% for the beam model. The arithmetic mean of the flap-wise and edge-wise errors are -2.0% for the shell model and -4.3% for the beam model.

3.2.2. Torsional response

The torsional twist is not directly available for the solid and shell models. An approximate method was developed that utilizes the displacement of four span-wise paths. This method was systematically applied to each of the model fidelities.

First, the undeformed model coordinates were obtained along the x_{LE} , x_{TE} , x_{HP} , x_{LP} span-wise paths defined in Fig. 9. The deformed coordinates of these curves are found by adding the respective displacements, u_{LE} , u_{TE} , u_{HP} , and u_{LP} as

$$\begin{aligned} X_{LE} &= x_{LE} + u_{LE} & X_{TE} &= x_{TE} + u_{TE} \\ X_{HP} &= x_{HP} + u_{HP} & X_{LP} &= x_{LP} + u_{LP} \end{aligned} \quad (1)$$

In general, each model fidelity resulted in their own displacement values.

A reference curve, r is defined as the arithmetic mean of x_{LE} , x_{TE} , x_{HP} , x_{LP} . At a given cross section, b_3 is then taken as the tangent of r . The vector pointing from x_{HP} to x_{LP} is found with $y_0 = x_{LP} - x_{HP}$. Since y_0 is not necessarily in the plane defined by b_3 , it is projected onto this plane with $y_1 = y_0 - y_0 \cdot b_3$.

Since each model likely predicts different relative motions of the spar caps, letting $y_1 = b_3$ was avoided. Instead, a second calculation was added to limit the effects of differing warping fields among the models. The vector pointing from trailing-edge to leading-edge was computed with $x_0 = x_{LE} - x_{TE}$. This vector was also projected onto the plane defined by b_3 as $x_1 = x_0 - x_0 \cdot b_3$. The desired vector is $y_2 = b_3 \times x_1$. An intermediate variable, y_3 , was defined as the arithmetic mean of y_1 and y_2 . b_2 was then taken as the Euclidean norm of y_3 . Finally, b_1 is simply $b_2 \times b_3$.

This process was repeated for every cross section and then also for the deformed coordinates to obtain B_1 , B_2 , and B_3 . The direction cosines relating the deformed configuration with the undeformed configuration is then $\beta_{ij} = B_i \cdot b_j$ (indicinal notation used here and only here). The three Euler angles, θ_1 , θ_2 , θ_3 are then obtained by standard means, where θ_3 is the torsional twisting angle plotted in Fig. 10(a). Subtracting off the solid model twist angles reveal the twist difference as shown in Fig. 10(b).

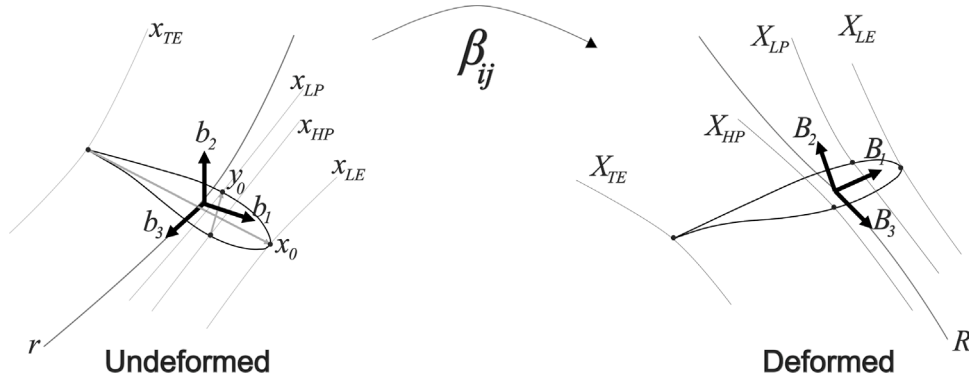


Fig. 9. Span-wise displacement paths and resulting coordinate bases in the deformed and undeformed configuration; used for determining torsional twisting angle for each model (beam, shell, solid).

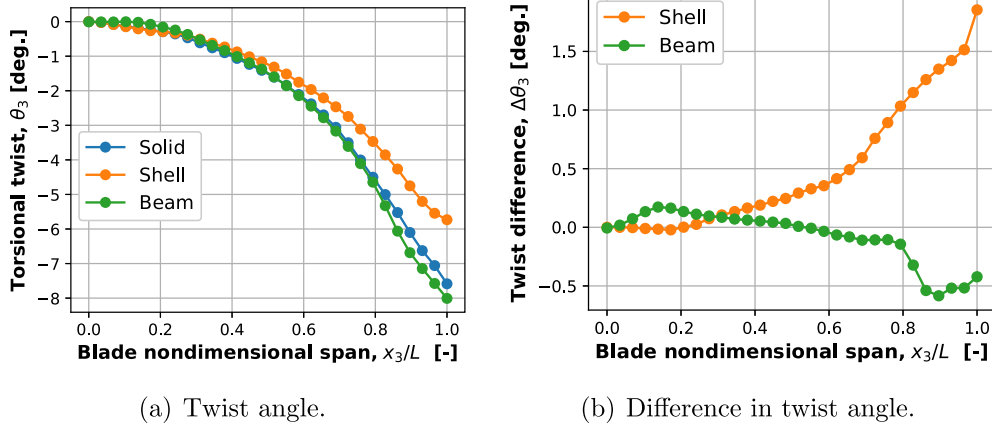


Fig. 10. Torsional angle comparison among each model fidelity.

It is notable that the beam model agrees very well with the solid model. In general, the torsional response near the root is slightly stiffer than the solid model and more compliant near the tip. The twist angle is only about 0.5° more compliant (or about 5.6%). In contrast, the shell model is generally stiffer throughout the blade span; yielding about a 1.5° deficit or (or about -24%).

It was noticed that the twist angle of the shell slightly reduces towards the blade tip. This artifact was traced to element size as illustrated in Fig. 11. Initially, the meshing scheme held fixed the nominal element size, l_e , throughout the blade length. The final shell meshing scheme fixed the number of elements about the cross sectional hoop direction and the refined mesh result is in Fig. 10 (and Tsai–Wu plots below). Note that the beam and solid models did not have this problem since thru-thickness element aspect ratios needed to be controlled.

3.2.3. Tsai–Wu failure index

Accurate calculation of stress or strain is needed when predicting material rupture. The complex nature of each of the six stresses (or strains) components creates difficulty in drawing meaningful comparisons between each model. Instead, comparisons were made with the 3D Tsai–Wu failure index since it was a concise measure that accounts for each stress component.

It is standard practice to neglect the out-of-plane stresses since they are expected to be much smaller in magnitude than the in-plane stresses. However, considering the poor out-of-plane strength that wind blade laminates have, a closer investigation is warranted. It is well known that shell models of wind blades generally result in very good in-plane stresses and poor out-of-plane stresses. For beam models, it is unknown how well the out-of-plane stresses are computed for anything

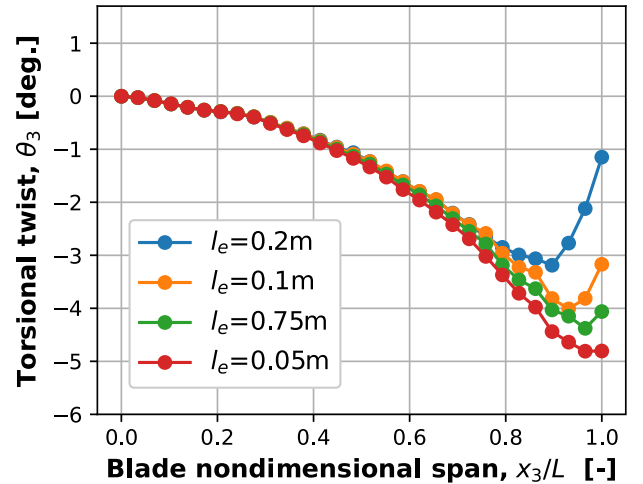


Fig. 11. Torsional blade twist of the shell model as the mesh is refined.

but the simplest of beams. Therefore, contributions to the 3D Tsai–Wu failure index are divided into in-plane and out-of-plane contributions to assess their relative importance.

The 3D Tsai–Wu failure index, f , is defined as a function of the six Cauchy stress components (σ_{ij})

$$f(\sigma_{11}, \sigma_{22}, \sigma_{12}, \sigma_{33}, \sigma_{23}, \sigma_{13}) = f_i(\sigma_{11}, \sigma_{22}, \sigma_{12}) + f_o(\sigma_{33}, \sigma_{23}, \sigma_{13}) \quad (2)$$

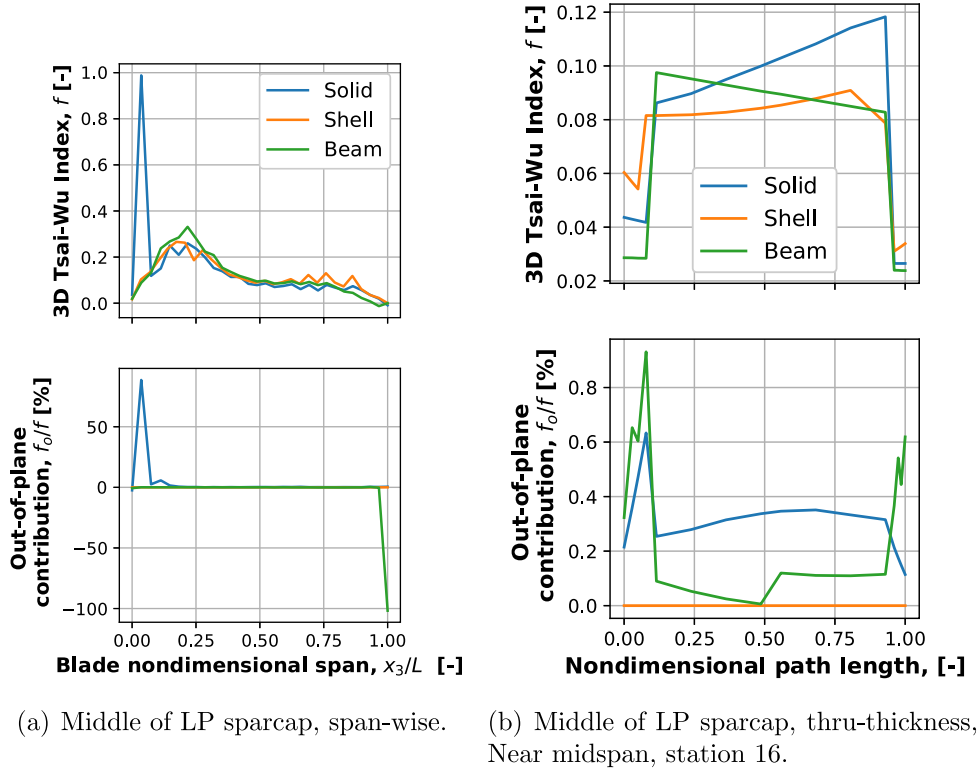


Fig. 12. Example paths with adequate agreement among all model fidelities.

where the contribution from the in-plane stresses is

$$f_i(\sigma_{11}, \sigma_{22}, \sigma_{12}) = F_1 \sigma_{11} + F_2 \sigma_{22} + F_{11} \sigma_{11}^2 + F_{22} \sigma_{22}^2 + 2F_{12} \sigma_{11} \sigma_{22} + F_{66} \sigma_{12}^2 \quad (3)$$

and that of the out-of-plane stresses is

$$f_o(\sigma_{33}, \sigma_{23}, \sigma_{13}) = F_3 \sigma_{33} + F_{33} \sigma_{33}^2 + 2F_{23} \sigma_{22} \sigma_{33} + 2F_{13} \sigma_{11} \sigma_{33} + F_{44} \sigma_{23}^2 + F_{55} \sigma_{13}^2 \quad (4)$$

F_i and F_{ij} are functions of the material tensile strengths, X , Y , Z , the compressing strengths, X' , Y' , Z' , and the shear strengths, S , R , T . Note that there is no guarantee that f , f_i , or f_o are always positive. For the adhesive material, $45/\gamma_\sigma$ MPa was used for the tensile and compressive strengths of the adhesive [38] where γ_σ is a factor of safety of 1.74. $20/\gamma_\sigma$ MPa was used for all of the shear strengths [38]. For the rest of the materials, the factored material strengths were obtained directly from [25]. Since the out-of-plane strengths (Z and Z'), were omitted in that publication, it was reasonable to set Z and Z' for each composite material to Y and Y' of the unidirectional glass laminate.

As described in Section 2.5, many sequences of coordinate points, called paths, were defined. Only the paths with noteworthy features are plotted in Figs. 12–16. In general, span-wise paths advance from the blade root to the blade tip while thru-thickness paths advance from the outer blade surface to the inner surface.

Fig. 12 shows a subset of paths where general agreement was found among all three models. Both Fig. 12(a) and (b) are located in the sparcap. Some paths outside of the sparcap also had adequate agreement but were omitted for brevity and because the failure index was insignificant in value. Note that the solid model is the only model that captures a failure index concentration near the root. This corresponds to the introduction of the carbon fiber sparcap. Fig. 12(a) also reveals that over 75% of the failure index concentration is comprised from the out-of-plane stresses.

Much like Fig. 12(a), Fig. 13 also illustrates a span-wise path where general agreement was found between each model. This path was in the aft web adhesive on the low-pressure side of the blade. Note that the shell model was omitted in this instance since the shell model in pyNuMAD does not account for web adhesives. Again, the solid model is exhibiting a large concentration of the failure index where the webs are first introduced. Not surprisingly, this geometric discontinuity is unaccountable with the beam model. It is notable that real wind blades gradually introduce the webs with the so-called *bird's mouth*. This effect was unaccounted for in any of the models since the pyNuMAD shell model lacks this capability. Had this effect been accounted for in the solid model, presumably the concentration shown Fig. 13 would be less pronounced.

Fig. 14 shows paths for which the beam model outperformed the shell model when compared to the solid model. The paths in Figs. 14(a) and 14(b) are located in the trailing-edge adhesive. They advance width-wise (rather than span-wise or thru-thickness) from the exterior of the blade towards the interior. The path in Fig. 14(a) was located in the 16th cross section and is representative of a midspan location. The beam and solid models show a concentration on the side of the adhesive closest to the blade interior. The beam model exaggerated the magnitude of the concentration while the shell model did not capture it at all. Furthermore, the overall trend from the shell model is completely deviated from the beam and solid models. The path in Fig. 14(b) was located in the second to last cross section and is representative of a cross section near the blade tip. The beam and solid model show that the adhesive is lightly loaded while the shell model largely over predicts the response.

Fig. 14(c) and Fig. 14(d) are thru-thickness plots near the edge of the spar cap. These paths begin in the outer facesheet followed by the carbon fiber sparcap. Then they pass through the interior facesheet followed by the web overwrap and finally through the web adhesive. Much of the response is captured by the beam model and hardly any is

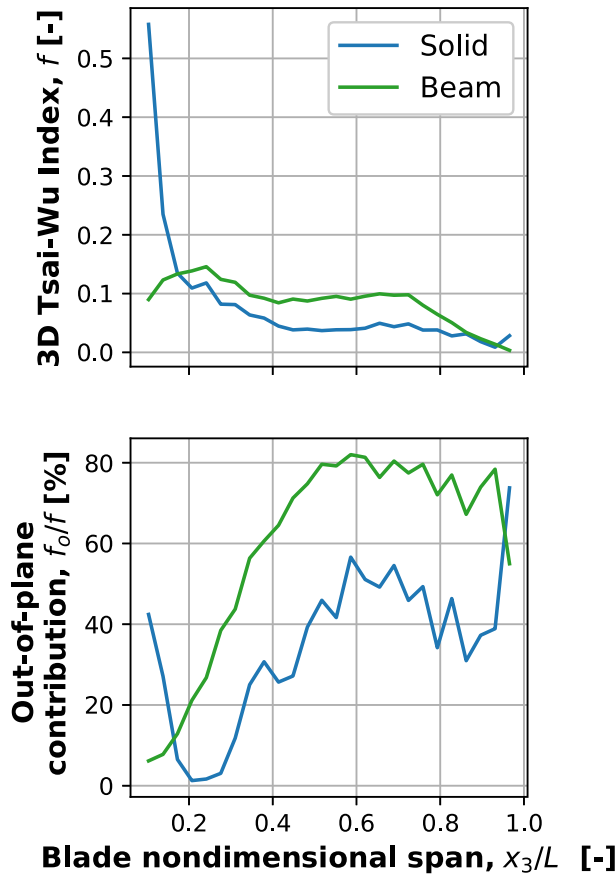


Fig. 13. Span-wise variation of the 3D Tsai-Wu failure index in one of the web adhesive connections.

captured by the shell model. It should be noted that the results from the shell model are ambiguous in their interpretation for this path. Because it lies on the joining point between the spar cap, panels, and shear web, there is discontinuity in the section definition and the interpolated results may contain uncertainty. The path in Fig. 14(c) is located near midspan. There, beam model seems to overshoot near the interior of the blade. This region of the path actually is in the web adhesive. The path in Fig. 14(d) is located near the blade tip. There the beam model captures the response near the blade interior but is largely inadequate in other parts of the path.

Fig. 15 shows paths for which the shell model outperformed the beam model when compared to the solid model. Both paths in Fig. 15 are thru-thickness paths and are near the leading-edge reinforcement. Fig. 15(a) is located in the root transition region whereas 15(b) in near midspan. It is remarkable that in both cross sections, the beam model response is null.

Fig. 16 shows paths for which neither the shell or beam models compare well with the solid model. The path location in Fig. 16(a) advances thru-the-thickness of the spar cap at the point of introduction. The solid model shows that the vast majority of the failure index is comprised of out-of-plane contributions. The shell and beam model are in good agreement with each other, and both show that there is negligible out-of-plane contributions. The path location in Fig. 16(b) advances thru-the-thickness of the foreweb; also at the point of introduction near the root. It is notable that the beam model does not register any values of significance. The shell model does but is over 50% under predicted.

3.3. Discussion

It was not surprising that the deflections between all the models agreed well. The shell model deflections stood out as being in even better agreement with the solid model. The beam model, however, was slightly more compliant than the shell and solid models. All three models account for geometric nonlinearity however, the beam model formulation is the only one that restricts the local rotations of material lines to be small. What this means is that effects such as the Brazier effect are not well accounted for. This is one possible explanation that could be explored further.

The torsional response of the beam model agreed with the solid model about as well as it did for the deflections. This was somewhat surprising for two reasons: (1) beam models are often referred to as low fidelity but in this case are outperforming the higher fidelity shell model. (2) a related study on a smaller blade found discrepancies [34], albeit, their solid models were layered-solid elements and here the facesheets and core layers are fully resolved. It seems that neglecting span-wise variations in blade topology (such as taper and initial twist flags in VABS) with the 2D cross sectional meshes is not adversely affecting torsional response.

The poor torsional response of the shell model was anticipated based on mixed results in the literature. Tavares et al. [39] shows that shell models are too torsionally compliant relative to a solid model. However, here and in another work [34] the shell response was overly stiff. Fluid-structure-interaction (FSI) simulations claim to offer high-fidelity responses of wind turbines. However, some FSI studies utilize shell models (eg [40,41]) which indicate a large likelihood of errors in the twist response; especially near the blade tip. Since the twist angle is directly related to aerodynamic angle-of-attack, results and conclusions from these models should be carefully examined.

Examination of Fig. 12(a), Fig. 13, and 16 reveals that only the solid model shows large concentrations of the Tsai-Wu failure index at span-wise geometric changes. Notably, these paths are located near the blade root. One location is where the spar cap is introduced and the other is where the web is introduced. This demonstrates the inadequacy of the beam and shell models to account for abrupt geometrical or material changes along the span. This was expected for the beam model since 2D cross sectional geometry makes accounting for the sudden introduction of the spar cap and webs an impossibility. However, to the authors' knowledge, this has never been quantified for beams since prior stress studies have been restricted to constant cross section idealizations [13]. It was more surprising that the shell model missed these concentrations since its geometry is indeed 3D and does model the sudden introduction of the spar cap and webs.

Fig. 14 showed two locations where the beam model outperformed the shell model by a large margin. One path location was at edge of the spar cap. The other path was in the trailing-edge adhesive. Both paths are recognized as locations with substantial discontinuity within the cross section. Poor shell model performance was expected here by recalling the geometric differences between the shell and the solid model (see 2.4). The beam model does better than the shell model but still is largely unpredictable. It appears that mechanical response in the trailing-edge adhesive can be sensitive to the way its shape and interaction with other components is defined.

Finally, Fig. 13 shows that besides the inboard and outboard edges of the webs, the beam model adequately captures the failure index. The figure also shows that the out-of-plane stresses have a significant role in the web adhesive connection. Even if the shell model had accounted for these adhesives, it is unlikely that it would produce reliable results due to the plane-stress assumption. Between this result and the poor trailing adhesive response discussed above, adhesive damage studies should avoid the use of shell elements.

The beam and shell model Tsai-Wu results have been shown to perform poorly at locations of abrupt geometric or material changes (both span-wise and chord-wise). Thus, failure predictions (cracking,

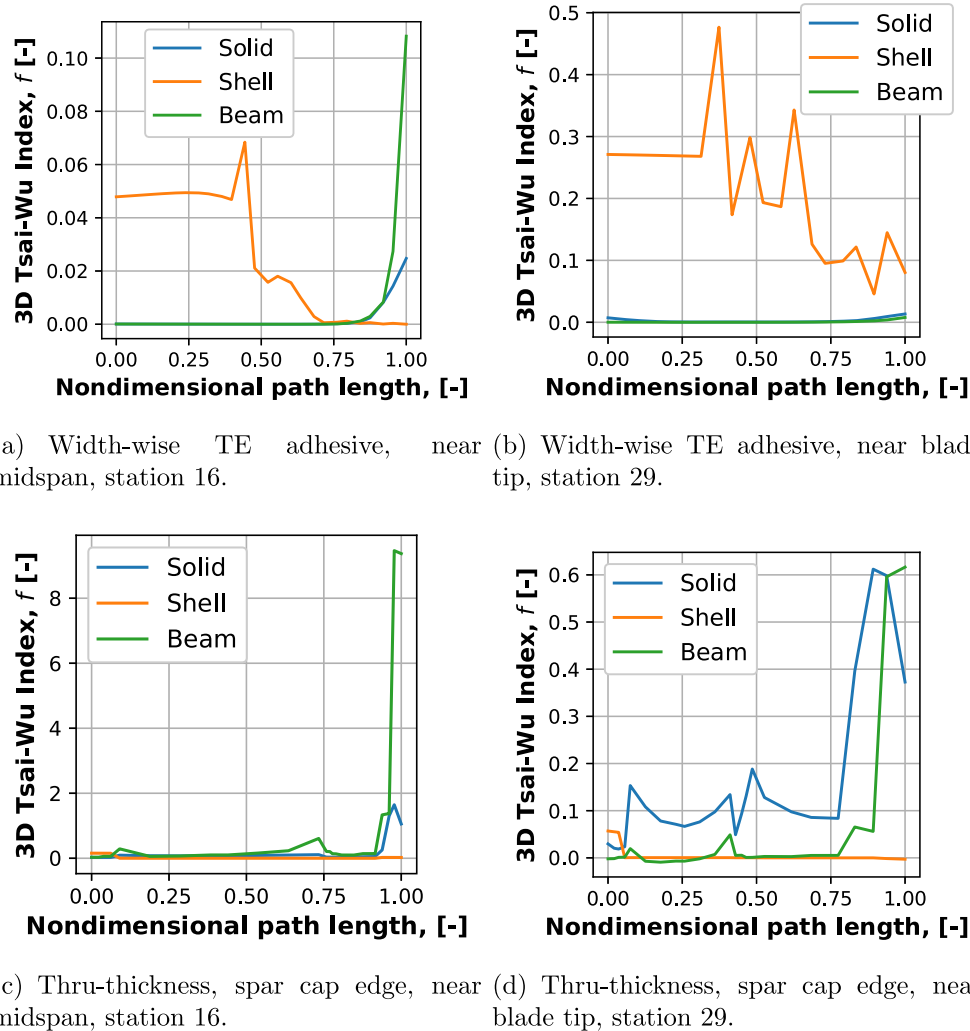


Fig. 14. Example paths where adequate agreement is only found between the solid and beam models.

delamination, rupture criteria, etc...) at trailing edges, web adhesives, ply-drops, spar-cap boundaries are unlikely to be accurate from beam or shell models. It is important to note that large out-of-plane stress contributions were encountered at these locations. Since layered-solid elements invoke similar assumptions to layered shells, it is unlikely that these finite elements will suffice.

4. Conclusion

This study presented a systematic comparison across beam, shell, and solid element models of a 100 m wind turbine blade. Deflections, torsional twist, and stresses were evaluated in the context of a static, large-deflection, flap-wise load. The literature considers the use of layered-solid elements as the highest fidelity. The distinctive feature here was the facesheet and core layers were explicitly discretized with at least two solid finite elements through-the-thickness. An unprecedented model size of 73.7 million elements revealed insights that have never been possible from prior experimental and numerical studies.

The global responses considered were deflections and torsional twist. All three model fidelities showed close agreement in transverse deflections. The tip response from the shell model was about 2% too stiff while that of the beam model was about double that (4.3%). The

beam model showed much closer agreement with the solid model in torsional twist angle, though the shell model showed some improvement with mesh refinement towards the blade tip. Shell models should be used with caution in aeroelastic studies, where the torsional response could have a significant impact.

The Tsai-Wu failure index was studied as a concise yet complete measure of the local stress response. Not surprisingly, all three model fidelities showed close agreement in most of the spar cap that were away from any discontinuities. The beam model showed notably better agreement with the solid model in the trailing-edge adhesive, which seems to be sensitive to the way its shape and interaction with other components is modeled. The shell model was in closer agreement at leading-edge. In general, both the beam and shell models were unsatisfactory in the prediction of out-of-plane stresses. These were found to be crucial at both span-wise and chord-wise layer transitions. Thus, failure predictions (cracking, delamination, rupture criteria, etc...) at trailing edges, web adhesives, ply-drops, spar-cap boundaries are unlikely to be accurate from beam, shell models, or layered-solid models (by extension). For these features, a high-fidelity solid model resolved with multiple elements through each layer is most reliable.

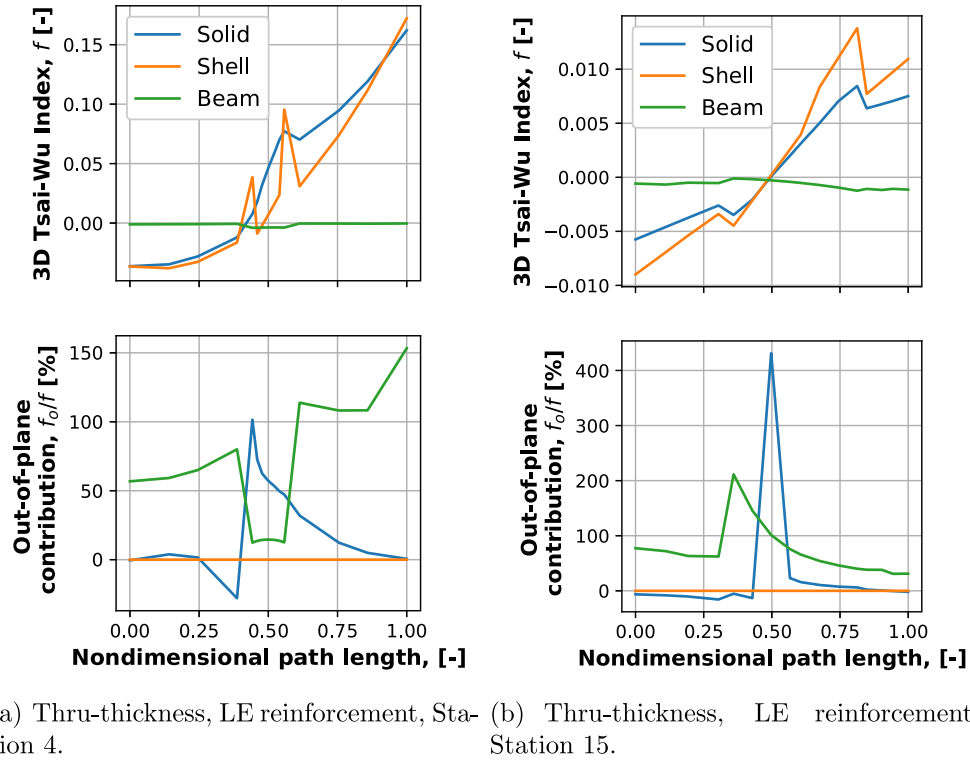


Fig. 15. Example paths where adequate agreement is only found between the solid and shell models.

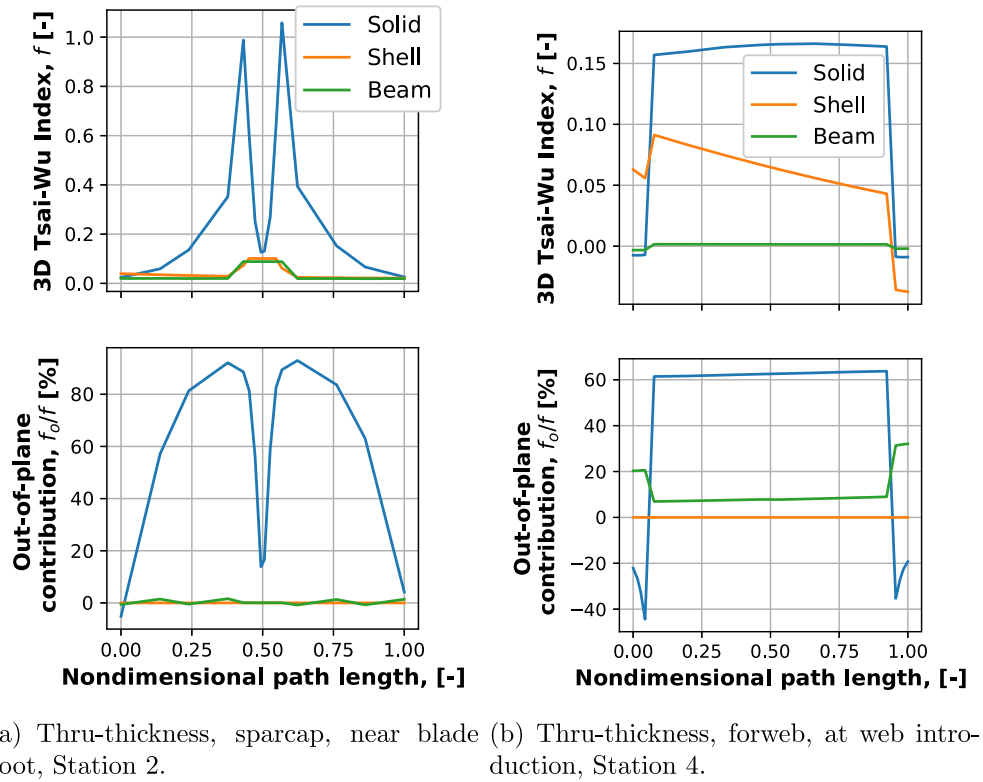


Fig. 16. Example paths where no adequate agreement is found with the solid model.

CRediT authorship contribution statement

Ernesto Camarena: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Evan Anderson:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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